9. The property which is characteristics of an Ionic compound is that

   a. it often exists as gas at room temperature.

   b. it is hard and brittle.

   c. it undergoes molecular reactions.

   d. it has low melting point.

10. Identify the following reactions as oxidation or reduction

    a. $Na \rightarrow Na^+ + e^-$

    b. $Fe^{3+} + 2\ e^- \rightarrow Fe^+$

11. Identify the compounds as Ionic/ Covalent/Coordinate based on the given characteristics.

    a. Soluble in non polar solvents

    b. Undergoes faster/instantaneous reactions

    c. Non conductors of electricity

    d. Solids at room temperature

12. An atom X with atomic number 20 combines with atom Y with atomic number 8. Draw the dot structure for the formation of the molecule XY.

13. Considering $MgCl_2$ as ionic compound and CH_4 as covalent compound give any two differences between these two compounds.

14. Why are Noble gases inert in nature?

III. Answer in detail

1. List down the differences between Ionic and Covalent compounds.

2. Give an example for each of the following statements.

   a. A compound in which two Covalent bonds are formed.

   b. A compound in which one ionic bond is formed.

   c. A compound in which two Covalent and one Coordinate bonds are formed.

   d. A compound in which three covalent bonds are formed.

   e. A compound in which coordinate bond is formed.

3. Identify the incorrect statement and correct them.

   a. Like covalent compounds, coordinate compounds also contain charged particles (ions). So they are good conductors of electricity.

   b. Ionic bond is a weak bond when compared to Hydrogen bond.

   c. Ionic or electrovalent bonds are formed by mutual sharing of electrons between atoms.

   d. Loss of electrons is called Oxidation and gain of electron is called Reduction.

   e. The electrons which are not involved in bonding are called valence electrons.

4. Discuss in brief about the properties of coordinate covalent compounds.

5. Find the oxidation number of the elements in the following compounds.

   a. C in CO_2

   b. Mn in $MnSO_4$

   c. N in HNO_3

REFERENCE BOOKS

1. Modern Inorganic Chemistry -R.D.Madan.

2. Textbook of Inorganic Chemistry -Soni, P.L. and Mohan Katyal.

INTERNET RESOURCES

1. https://youtu.be/G08rZ6xiIuA

2. https://youtu.be/LkAykOv1foc

3. https://youtu.be/DEdRcfyYnSQ

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